

Software Requirements Specification (SRS)

Hospital Booking System for the Outpatient Department (OPD) | Team Aura | Version 1.0 - revised draft

Document Control

Field	Value
Team name	Team Aura
Project title	The Hospital Booking System
Version	1.0 -revised draft
SDLC	Incremental / Iterative (or Agile)
GitHub repository	https://github.com/jazzvarichsara-stack/The-hospital-booking-system

1. Introduction

1.1 Purpose and Scope

The purpose of the Hospital Booking System for the Outpatient Department (OPD) is to allow patients to view doctor information, specializations and available schedules, book appointments online, receive token numbers, and check the current token queue. The system also supports administrators in managing doctors, schedules, appointments, and patient queues.

1.2 Users and Stakeholders

Role	Type	Needs / Responsibilities
User (Patient)	Primary system user	Register/login, view doctor information and availability, book appointments, receive a token number, check the current queue, and cancel or reschedule appointments.
Admin	System role	Manage doctor information, schedules, patient appointments, token numbers and queues; call the next patient and update token status.
Doctor	Secondary stakeholder	Needs accurate schedule and appointment information. The current system does not define a doctor-facing role.

2. Overall Description

2.1 Product Context

The product is a web-based system for outpatient appointment booking and queue management. The defined system roles are User (Patient) and Admin.

Patient flow: register/login → view doctor information and availability → select a doctor, date, and time → book an appointment → receive a token number → check the current queue. Admins manage doctor information, schedules, appointments, and token queues; they can call the next patient and update token status.

2.2 In-Scope Functions

- FR-1: User Registration and Login.
- FR-2: View Doctor Information and Availability.
- FR-3: Book an Appointment.
- FR-4: Check Current Token Queue.
- FR-5: Cancel or Reschedule an Appointment.
- FR-6: Manage Doctors, Schedules, Appointments, and Token Queue.

2.3 Out-of-Scope Functions

- Online payment or billing.
- Online medical consultation or video calls.
- Medical diagnosis or treatment recommendations.
- Prescription management and electronic medical records management.
- Emergency or ambulance services.
- Insurance claim processing.
- Online pharmacy or medicine ordering.
- Integration with external hospital systems.
- AI-based medical diagnosis.

2.4 Assumptions and Constraints

- Patients can access the system through the internet using an electronic device.
- The Admin keeps doctor information, specializations, department assignments, and available schedules up to date.
- Users provide personal information when registering and making appointments.
- Users must have an account and be logged in before booking, changing, or cancelling an appointment.
- Only Admins can add, edit, or remove doctor profiles.
- Only Admins can call patients and update token status.
- Every confirmed appointment receives a system-generated token number.
- Token statuses are Waiting, Called, In Operation, Completed, and Cancelled.
- The system supports outpatient booking and queue management; it does not provide diagnosis, treatment advice, or emergency support.

3. Functional Requirements

FR-1 — User Registration and Login

The system shall allow patients to create an account and sign in before using appointment and queue-related functions. Queue and appointment information shall be associated with the relevant patient account.

Acceptance criteria: Given valid registration information, the system creates a patient account. Given valid credentials, the system allows the user to sign in. Invalid credentials do not grant access.

FR-2 — View Doctor Information and Availability

The system shall allow patients to view doctors, their specializations, and available appointment dates and times.

Acceptance criteria: When the patient opens the doctor list, the system displays doctor information and specializations. When a doctor is selected, the system displays available dates and times. Unavailable times cannot be selected for booking.

FR-3 — Book an Appointment

The system shall allow patients to select a doctor, date, and available time and confirm an appointment online.

Acceptance criteria: When the patient confirms an available slot, the system creates the appointment. If the slot is already taken, the system prevents the booking. After successful booking, the system generates a token number.

FR-4 — Check Current Token Queue

The system shall allow patients to view their token number, the current queue, and the status of their appointment.

Acceptance criteria: For a patient with a valid appointment and token, the system displays the current token number and the patient's token number. The system displays the updated token status, including Completed or Cancelled when applicable.

FR-5 — Cancel or Reschedule an Appointment

The system shall allow patients to cancel or reschedule an upcoming appointment.

Acceptance criteria: When a patient cancels an upcoming appointment, its status changes to Cancelled. When a patient selects an available date and time, the appointment is rescheduled. The system prevents selection of an unavailable reschedule time.

FR-6 — Manage Doctors, Schedules, Appointments, and Token Queue

The system shall allow Admins to manage doctor information, schedules, patient appointments, and token queues.

Acceptance criteria: When Admins add, edit, or remove doctor information, the system records the changes and reflects updated information to patients. Admins can review appointments and update appointment status. The system displays queue order and removes cancelled tokens from the active waiting list. Admins can call the next patient and update token status.

4. Non-Functional Requirements

ID / Type	Requirement	Verification
NFR-1 Performance	Doctor and appointment schedule pages shall load within 3 seconds under normal network conditions.	Measure page load time under normal network conditions.
NFR-2 Performance	The system shall create and display a patient's token number within 2 seconds after booking confirmation.	Measure the time between booking confirmation and token display.
NFR-3 Usability	A first-time user shall be able to find a doctor and complete an appointment booking in under 2 minutes without assistance.	Conduct a timed trial with a first-time user.
NFR-4 Reliability	The system shall save a confirmed appointment and its token number so that the records remain available after the user leaves or refreshes the appointment page.	Create an appointment, refresh the page, and verify the appointment and token records.
NFR-5 Security / Privacy	A user shall only be able to view and manage their own personal information and booking history.	Log in as one user and attempt to access another user's information; access must be prevented.

5. Use Cases

Use Case	Actor	Related FR	Brief Success Outcome
UC-1 Register and log in	User (Patient)	FR-1	The patient creates an account or signs in.
UC-2 View doctor information and availability	User (Patient)	FR-2	The patient views doctors, specializations, and available schedules.
UC-3 Book an appointment and receive token	User (Patient)	FR-3	The patient books an available slot and receives a token number.

UC-4 Check current token queue	User (Patient)	FR-4	The patient views the current queue, their token number, and token status.
UC-5 Cancel or reschedule an appointment	User (Patient)	FR-5	The patient cancels or reschedules an upcoming appointment.
UC-6 Manage doctors, schedules, appointments, and queue	Admin	FR-6	The Admin updates doctor/schedule information, manages appointments, calls the next patient, and updates token status.

6. Token Status Model

The token statuses specified in the project materials are Waiting, Called, In Operation, Completed, and Cancelled. The exact token-number assignment algorithm (for example, the ordering rule) has not been defined in the available materials and requires team confirmation.

7. Traceability Summary (Golden Thread)

Problem / Need	Requirement	Screen IDs	Test focus
Patients need to find doctors and check availability.	FR-2	S-03, S-04	Doctor list, schedule display, unavailable slot.
Patients need to book appointments and receive a token.	FR-3	S-05, S-06	Successful booking, token generation, duplicate slot prevention.
Patients need to know queue progress.	FR-4	S-07	Current queue, patient token, status update.
Patients may need to change an appointment.	FR-5	S-08, S-09	Cancellation and rescheduling with unavailable slot.
Admins need to maintain accurate doctors, schedules, appointments, and queues.	FR-6	S-10–S-14	Admin edits, appointment status, queue order, call next, token status.